

Marco Maida

Senior Machine Learning Engineer

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I'm a software engineer with over **ten years of experience** building systems that address complex, real-world problems. I worked in the fields of **industrial software**, **videogames**, **real-time systems**, **formal verification**, and **autonomous vehicles**. I'm currently based in **London**, working as a **Senior Machine Learning Engineer** at **Wayve**. I sit between model development and production inference, co-designing models and runtime architectures to run Wayve's driving models efficiently on resource-constrained accelerators. I work mainly in **C++**, **Rust**, and **Python**, and I'm particularly interested in problems where **performance**, **correctness**, and **practical constraints** come together.

Professional Experience

Wayve Ltd

London, United Kingdom

- **Senior Machine Learning Engineer**

May 2026–Present

I work in inference optimisation, between model training and inference. I co-design models and runtime architectures around hardware constraints, partition workloads across CPUs, GPUs, and NPUs, and develop execution strategies that process model components as soon as their sensor inputs become available. I optimise end-to-end latency, throughput, resource utilisation, and numerical fidelity. (*C++*, *Rust*, *Python*, *accelerators*).

- **Senior Software Engineer**

Dec 2022–May 2026

I designed and built Wayve's real-time inference pipeline, initially independently and later as tech lead of a team of four. The platform powers every Wayve vehicle and its simulation environments, supports multiple inference frameworks and accelerators, and serves both real-world validation and production driving. Earlier, I built key on-board sensor, data-collection, upload, and vehicle-networking components. (*ROS2*, *Linux*, *C++*, *Rust*, *Python*).

2016–2019 **Software Engineer – Videogames**

34BigThings

I contributed to five major game titles, focusing on game infrastructure, AI, gameplay, and development tools. I worked on both single-player and online multiplayer games shipped on Steam, PS4, XboxOne, Switch, and mobile platforms. (*Unity3D*, *C#*, *Unreal Engine*, *C++*).

2015–2016 **Software Engineer – Videogames, Simulation**

Maserati, Choralia

I led two freelance B2B projects: developing an educational game for mobile and browsers, and collaborating to create a 3D visualization tool for product presentation. (*Unity3D*, *C#*, *JavaScript*).

2013–2016 **Software Engineer – Industrial Software**

R.O. srl

I contributed to a suite of software solutions for glass processing factories, focusing on order tracking and optimizing machine work, product waste, and logistics. During my third year, I managed a team of 4 junior engineers. (*C*, *C++*, *C#*, *SQL*).

Education and Research

2022 **Research Internship**

Bloomberg LP

I worked on accelerating SAT solving using GPUs. (*C++*, *CUDA*).

2019–2022 **PhD Student**

Max Planck Institute

My research focused on formal verification and real-time systems, specifically verifying the timeliness of software systems. I published three papers and mentored three interns. (*COQ*, *C*, *Rust*).

2019–2022 **Master's in Computer Science**

Technische Universität Kaiserslautern

I specialized in real-time systems as part of a joint master's and PhD program.

2016–2019 **Bachelor's in Computer Science**

Università degli Studi di Torino

I specialized in computability and formal methods.

Selected Publications

- 2025 **Claycode: Stylable and Deformable 2D Scannable Codes** SIGGRAPH 2025
★ *Journal publication in Transactions of Graphics. Selected for CAF trailer.*
We introduced Claycodes, a new type of visual code that breaks free from traditional rigid grids like QR codes. Claycodes encode data as a tree structure and allow full customizability and deformation, with real-time scanning capabilities. (<https://arxiv.org/abs/2505.08666>).
- 2022 **From Intuition to Coq: A Case Study in Verified Response-Time Analysis, FIFO Scheduling** RTSS 2022
We developed a formally verified response-time analysis for FIFO schedulers, challenging traditional pen-and-paper methods. (<https://people.mpi-sws.org/~kbedarka/rtss22.pdf>).
- 2021 **Foundational Response-Time Analysis as Explainable Evidence of Timeliness** ECRTS 2022
★ *Outstanding Paper Award.*
I developed POET, a tool for formally verified worst-case scenario timing analysis. (<https://drops.dagstuhl.de/opus/volltexte/2022/16336/pdf/LIPIcs-ECRTS-2022-19.pdf>).

Selected Open-Source Projects

- 2025 **Claycode** <https://claycode.io>
Inventor and lead contributor. Claycode includes a generator website, an Android scanner app, and a large amount of research code.
- 2020-2022 **PROSA** <https://gitlab.mpi-sws.org/RT-PROOFS/rt-proofs>
Main contributor to PROSA, one of the leading formally-verified frameworks in the real-time systems community.
- 2022 **Treecode** www.maida.me/treecode
The precursor to Claycode, a novel 2D scannable code that encodes messages as unique trees.
- 2021 **POET** <https://gitlab.mpi-sws.org/RT-PROOFS/POET>
First-ever implementation of a foundational response-time analysis tool, created as part of my first academic publication.
- 2018 **Fast Mobile Cycle (FMC) Framework and Toolkit** www.github.com/340openThings
Developed a Unity3D framework that accelerates production-ready casual game creation, paired with a Python toolkit for bulk operations.